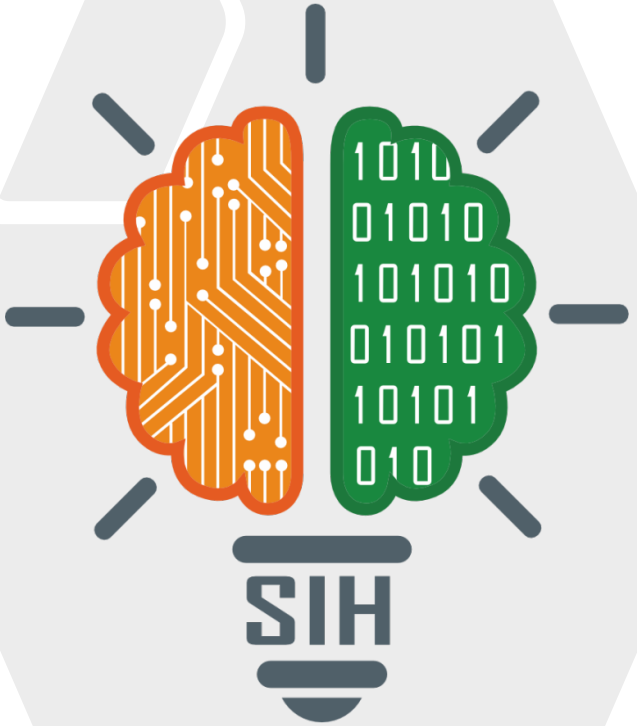
SMART INDIA HACKATHON 2026

SMART INDIA HACKATHON 2026

NidhiSetu

- **Problem Statement ID – SIH26265**
- **Problem Statement Title – Streamline of security deposit/ FDR portal.**
- **Theme – Smart Automation**
- **PS Category – Software**
- **Team ID – [Team ID]**
- **Team Name (Registered on portal) – [Team Name]**



[Team Name]

NidhiSetu

SMART INDIA HACKATHON 2026

Every security deposit confirmed with the bank, and tracked from submission to release

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

Detailed explanation of the proposed solution

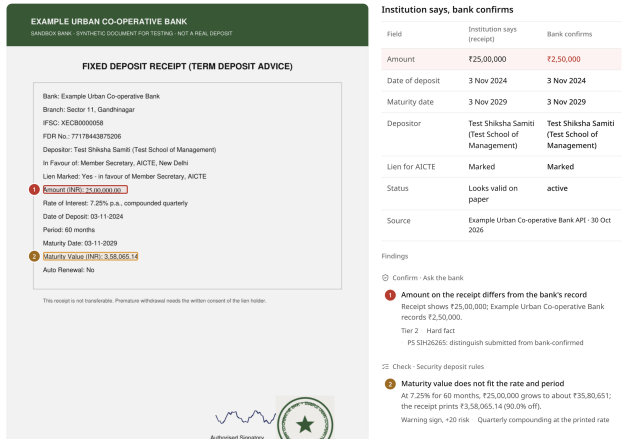
- Reads each FDR or e-FDR: amount, bank, **IFSC**, dates, lien; **OCR** for phone photos
- Checks the **deposit rules**: in favour of AICTE, lien, amount, tenure, branch
- **Asks the bank**: e-FDR signature, term-deposit API, or a signed branch link
- **Tracks the lifecycle**: reminders at 90/60/30/7 days, escalation, renewal
- **Release workflow**: conditions checked, two officials, discharge letter to the bank

How it addresses the problem

- A deposit today is a paper receipt checked once. In Chandigarh, **₹116.84 crore** of FDRs the bank had never issued went unnoticed **for over a year** (Tribune, Mar 2026)
- NidhiSetu shows **what the institution says next to what the bank confirms**, for every FDR, on day one
- A **bank re-check every 90 days** finds an early closure in about **45 days**, not at maturity

Innovation and uniqueness of the solution

- **Bank-confirmed, not paper-trusted**: 3 tiers; never uses a QR or phone printed on the receipt
- **Continuous re-confirmation**, so a closed deposit cannot hide
- **Pledge registry**: one FDR cannot secure two institutions
- **Tamper evidence**: white patches, font changes, maturity-value arithmetic
- **Maker-checker release**, hash-chained audit trail



Working prototype: the receipt says ₹25 lakh, the bank says ₹2.5 lakh; numbered boxes mark the evidence

80/80 forgeries caught **0/60** false alarms

0.14 s per receipt on a laptop CPU · 140 labelled test receipts

1 READ
Every field off the receipt or e-FDR, with its place on the page: bank, IFSC, number, amount, dates, lien. OCR for phone photos

›

2 CHECK
Security deposit rules: in favour of AICTE, lien marked, amount and tenure enough, branch on RBI's list, not pledged twice

›

3 CONFIRM
Ask the bank, never the paper: e-FDR signature, term-deposit API, or a signed one-time link to the branch

›

4 DETECT
Tampering signs: white patch over text, a different font on a key value, maturity value that does not fit the rate

›

5 MONITOR
Every day: reminders at 90, 60, 30, 7 days, escalation when overdue, bank re-check every 90 days

›

✓ RELEASE
Conditions checked, two officials approve, the bank is asked to remove AICTE's lien. Every step on record.

Demo video: [add unlisted link] **Live prototype:** [add link] **Code:** [add GitHub link]

[Team Name]

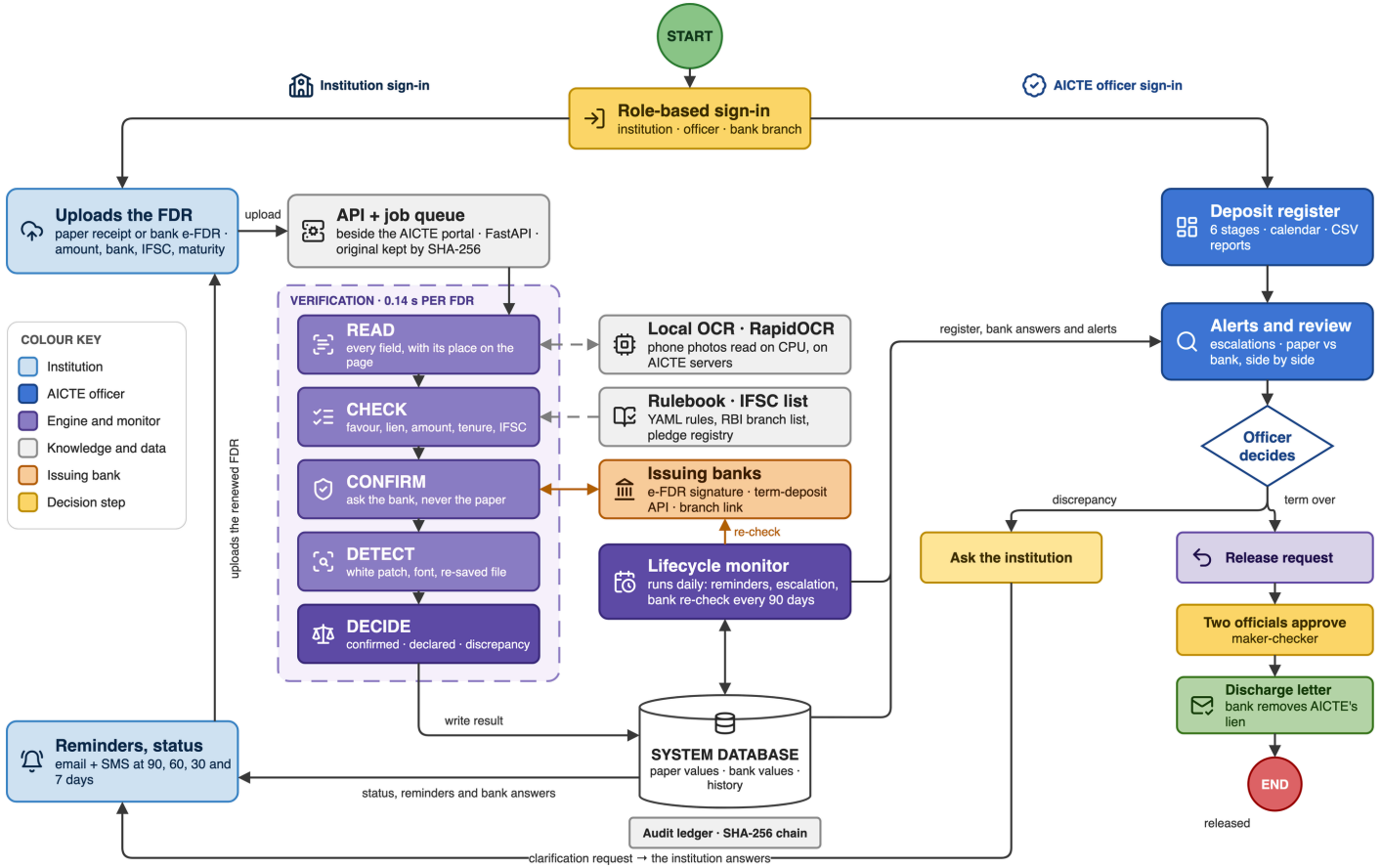
TECHNICAL APPROACH

SMART INDIA HACKATHON 2026

Technologies to be used (e.g. programming languages, frameworks, hardware)

- **FastAPI · Python 3.12**: typed API; Python owns the PDF, OCR and signature libraries (vs Node.js: weak PDF tooling)
- **PostgreSQL 16**: every instrument with paper and bank values side by side; demo runs the same code on SQLite (vs MongoDB: weak transactions for financial records)
- **Redis + RQ workers**: the daily monitor and bank re-checks run as jobs (vs Celery: heavier; Kafka: overkill)
- **MinIO (S3) on NIC / MeghRaj**: receipts stored by SHA-256, encrypted (vs local disk: does not scale)
- **pyHanko**: verifies the bank's PAdES signature on an e-FDR; catches edits after signing
- **RapidOCR (PP-OCR, ONNX)**: reads phone photos on a CPU (vs cloud OCR: financial data leaves AICTE)
- **Bank channel adapters**: e-FDR, term-deposit API, signed branch link; Account Aggregator next (vs letters to branches: weeks, no trail)
- **RBI IFSC list + rapidfuzz**: the branch must exist; OCR-tolerant name matching (vs exact match: photos fail)
- **React 18 + TypeScript on UX4G 3.0**: the Government of India design system, accessibility bar (vs a custom look)
- **Role-based sign-in + SHA-256 hash chain**: maker-checker roles; tamper-evident log (vs blockchain: same proof, no cost)

Methodology and process for implementation (Flow Charts/Images/ working prototype)



100% open source · ₹0 licence · CPU only, no GPU

1 Built lifecycle portal, bank sandbox, monitor, 6 passing tests

2 Measured 140 labelled receipts, 8 forgery patterns

3 Pilot our college's deposit records, then one regional office

[Team Name]

FEASIBILITY AND VIABILITY

SMART INDIA HACKATHON 2026

Analysis of the feasibility of the idea

Technical

Built and running today; banks already issue e-receipts and share term deposits through AA

Economic

₹0 licence; replaces verification letters and follow-up calls; NIC cloud

Operational

Fits the Regulation Bureau's steps; officers keep every decision

Legal & data

On-prem, role-based, encrypted; bank checks only where authorised

80/80
forgeries caught

0/60
false alarms (20 phone photos)

0.14 s
per receipt, laptop CPU

45 days
to catch an early closure (paper: 1,178)

Measured on 140 synthetic receipts (8 patterns from reported cases) and a 10,000-deposit simulation; the pilot measures real-world rates.

CONFIRM: ask the bank that issued it

TIER 1 instant — **Bank-signed e-FDR**
The bank's digital signature, checked against its registered certificate; a QR code counts only if it opens the bank's allow-listed domain

TIER 2 seconds — **Bank's term-deposit API**
Amount, dates, AICTE's lien and status, straight from the bank's records. Asked again every 90 days, so an early closure cannot hide

TIER 3 days — **Signed branch confirmation**
A one-time "Did your branch issue this?" link to the branch contact held in the registry. No answer within the SLA escalates to the Regional Officer

Never trust a QR code, phone number or email printed on the receipt itself. The bank's record decides; the paper only proposes.

EVERY DEPOSIT SITS IN ONE LIFECYCLE STAGE (WORDING FROM THE PS)

ACTIVE | EXPIRING | MATURED | RENEWED | RELEASED | PENDING

Potential challenges and risks — Strategies for overcoming these challenges

Challenge or risk	Strategy
Bank has no API (a small co-operative bank)	Tier 3 signed branch link; no answer within the SLA escalates; AA consent as banks join
Bank service down or slow	Stays "awaiting bank", never "fake"; retried automatically
Genuine receipt flagged (phone photo, OCR)	OCR-tolerant matching: 0/20 false alarms on photos; officer sees the evidence first
FD closed after it was submitted	Bank re-check every 90 days; lien must show at the bank, not only on paper
A clever PDF edit	The bank's record decides, not the file; forensics is a second signal
Rules or amounts change	Rulebook in YAML, versioned per cycle; no code change

Viability

- **Pilot**: our college's deposit records (with permission), then one AICTE regional office, then national
- **Reuse**: UGC, NCTE, PCI and any department that holds FDRs as security, like the municipal bodies in the 2026 cases
- **Sustainable**: open source under AICTE; each bank is onboarded once, through an adapter
- **Scale estimate**: if AICTE holds ~20,000 deposits, quarterly re-checks are ~220 bank calls a day; one worker does that in under a minute

[Team Name]

IMPACT AND BENEFITS

SMART INDIA HACKATHON 2026

Potential impact on the target audience

R — AICTE Regulation Bureau
One register for every deposit; reminders and escalations automatic; release with a full trail

I — Honest institutions
Upload once, get reminded before maturity, get the deposit released without chasing letters

B — Banks
Answer once through an API or a one-click link, instead of letters and phone calls

S — Students and the public
The security behind each approved institution is real money at a bank, not a piece of paper

Benefits of the solution (social, economic, environmental, etc.)

Social

- Deposits that back approvals are real
- Faster, transparent release for genuine institutions
- Fewer disputes: every step on record

Economic

- A fake or closed FDR is caught in weeks, not years
- Officer time saved on letters and follow-up
- ₹0 licence, no per-check fee

Environmental

- Paperless from upload to discharge letter
- No couriered receipts or verification letters
- Bank e-FDRs preferred over paper

Governance

- Institution's claim and bank's record kept apart, as the PS asks
- Two officials on every release
- Tamper-evident audit trail

Why it matters: reported cases

₹116.84 Cr
of Chandigarh MC's FDRs were not in the bank's system; nobody noticed for over a year (The Tribune, Mar 2026)

₹150 Cr
of Panchkula MC's fixed deposits did not exist; found only on a maturity transfer request (Moneylife, Mar 2026)

₹64 Cr
swindled from Telugu Akademi's fixed deposits using forged FDRs and letters (Moneylife, 2021)

IFN 760 COV
Bank guarantees travel bank-to-bank on SFMS, the only valid mode (PNB). NidhiSetu gives FDRs a bank-side check

[Team Name]

RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2026

Details / Links of the reference and research work

How NidhiSetu compares with what exists today

	Paper file + letters	Spreadsheet tracker	NeSL e-BG	Account Aggregator	NidhiSetu
Bank confirms the deposit	Letters	No	BGs only	With consent	Yes
Covers FDRs held as security	Yes	Yes	No	Data only	Yes
Reminders before maturity	No	Manual	n/a	No	Yes
Catches an early closure	At maturity	No	n/a	If re-fetched	Every 90 d
Two-official release, audit trail	Partly	No	n/a	No	Yes
Paper vs bank, side by side	No	No	n/a	No	Yes

What exists, and what we build on

- **NeSL e-BG**: banks already issue bank guarantees electronically with instant verification; we bring that idea to FDRs
- **Account Aggregator**: term deposits are a listed data type; our next bank channel, with the institution's consent
- **RBI IFSC lists** (open dataset): every branch on a receipt must exist
- Stakeholder input: [add one line from your college's accounts officer or AICTE coordinator]

Research, standards and sources

1. PS SIH26265, AICTE Regulation Bureau (sih.gov.in)
2. AICTE, Simple steps for making security deposit online
3. The Tribune, Chandigarh MC: Rs 116 crore FDR scandal unfolds, 11 Mar 2026
4. Moneylife, ₹150 crore FD discrepancy in Panchkula MC account, Mar 2026
5. Business Today, CBI arrests ex-bank officials in Haryana FD fraud case, 30 Jun 2026
6. Moneylife, ED to probe Rs64 crore Telugu Akademi fixed deposit fraud
7. PNB, Bank guarantees transmitted through SFMS IFN 760 / 760 COV
8. CSB Bank, Electronic bank guarantee through NeSL
9. Sahamati, Account types activated by banks on AAs (term deposits)
10. Sahamati, Glossary of AA terms (TD: term / fixed deposit)
11. Razorpay IFSC dataset, built from RBI NEFT/RTGS lists
12. ETSI EN 319 142 (PAdES): PDF digital signatures
13. Du et al., PP-OCR: a practical ultra lightweight OCR system, arXiv:2009.09941
14. UX4G 3.0, Government of India design system
15. GIGW 3.0: Indian government web guidelines (WCAG 2.1 AA)

Data and test assets (all free)

- **Labelled test set we built**: 140 receipts, 8 forgery patterns from the cases above
- **Sandbox banks**: our own CA for signed e-FDRs, a term-deposit API, branch links
- **10,000-deposit simulation** of early closures: paper check vs 90-day re-check
- **Rules**: AICTE security deposit norms, configurable per cycle (PS allows synthetic data)